

SYSTEM DESIGN DOCUMENT

Architecture Style

Layered Monolithic Architecture

Frontend (React)



REST API Layer



Service Layer



Repository Layer



MySQL Database

High Level Architecture

React Frontend (Vercel)



Axios



Spring Boot REST APIs (Render)



Spring Security + JWT



Service Layer

↓

JPA Repositories

↓

MySQL (Railway)

↓

Cloudinary (File Storage)

Backend Folder Structure

src/main/java/com/teamsync

├── config

├── security

├── controller

├── service

├── repository

├── entity

├── dto

├── mapper

├── exception

├── util

└── TeamsyncApplication.java

config

- SecurityConfig
- CorsConfig
- CloudinaryConfig

security

- JwtFilter
- JwtService
- CustomUserDetailsService

controller

- AuthController
- UserController
- ProjectController
- TaskController
- CommentController
- NotificationController
- AnalyticsController

service

- AuthService
- UserService
- ProjectService
- TaskService
- CommentService
- NotificationService
- AnalyticsService

repository

- UserRepository
- ProjectRepository
- TaskRepository
- CommentRepository
- NotificationRepository

Security Flow

User Login



Validate Credentials

↓

Generate JWT

↓

Return Token

↓

Frontend Stores Token

↓

Protected API Request

↓

JWT Filter Validates Token

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Request Continues

Role Authorization

ADMIN

Can:

- Manage Users
- Manage Projects
- Access Reports
- Access Analytics

LEAD

Can:

- Manage Assigned Projects
- Create Tasks
- Assign Tasks
- View Team Analytics

MEMBER

Can:

- View Assigned Tasks
- Update Task Status
- Add Comments

File Upload Flow

User Uploads File



Backend Receives Multipart File



Upload To Cloudinary



Cloudinary Returns URL



Save URL In Database



Return URL To Frontend

Notification Flow

Task Assigned



Create Notification Record



Store In Database



Show In Notification Center

Dashboard Data Flow

Frontend Dashboard



Analytics API



Aggregate Queries



Response DTO



Charts + Metrics

Future Enhancements

Phase 2

- Redis Caching
- WebSockets
- Real-Time Notifications
- Kanban Board
- Activity Streaming

Phase 3

- AI Task Suggestions
- GitHub Integration
- Google Calendar Integration
- Team Chat

Deployment Pipeline

GitHub



Vercel (Frontend)

↓

Render (Spring Boot Backend)

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Railway MySQL

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Cloudinary

Monitoring

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Sentry